Rushikesh Patil

2124UCSM1037

command-line interface (CLI)

Commands for Shell Programming / CLI

A command-line interface (CLI) is a text-based way to interact with the operating system by typing commands. In shell programming, commands are used to perform file operations, directory handling, process control, and other system tasks.

Common CLI Commands

1. pwd
Displays the present working directory.
Example:

```
Pwd
```

2. ls
Lists files and directories.
Example:

```
ls
ls -l
ls -a
```

3. cd
Changes the current directory.
Example:

```
cd Documents
cd ..
```

4. mkdir
Creates a new directory.
Example:

```
mkdir myfolder
```

5. rmdir
Removes an empty directory.
Example:

```
rmdir myfolder
```

6. touch
Creates a new empty file.
Example:

```
touch file1.txt
```

7. cat
Displays the content of a file.
Example:

```
cat file1.txt
```

8. cp
Copies files or directories.
Example:

```
cp file1.txt file2.txt
```

9. mv

Moves or renames files and directories.

Example:

```
mv file1.txt newfile.txt
```

10. rm

Removes files or directories.

Example:

```
rm file1.txt
rm -r myfolder
```

11. echo

Displays text on the screen or writes text into a file.

Example:

```
echo Hello
echo "Hello" > file.txt
```

12. whoami

Displays the current username.

Example:

```
Whoami
```

13. date

Shows current system date and time.

Example:

```
date
```

14. clear

Clears the terminal screen.

Example:

```
Clear
```

15. man

Displays the manual of a command.

Example:

```
man ls
```

Basic Shell Programming Commands

16. Variable Declaration

```
name="Rushikesh"
echo $name
```

17. Taking User Input

```
read name
echo $name
```

18. Conditional Statement

```
if [ $a -gt $b ]
then
echo "A is greater"
fi
```

19. Loop

```
for i in 1 2 3
do
echo $i
done
```

20. Comments

```
# This is a comment
```

Conclusion

Shell programming and CLI commands help users interact with the operating system efficiently. These commands are useful for file management, directory handling, automation, and writing shell scripts.